6. Project Testing

FitBuddy — Test strategy, current test cases, expected outcomes, and quality checks

Testing Strategy

The supplied test suite uses FastAPI TestClient and exercises the application through HTTP-level behavior. This is appropriate for verifying that routing, template rendering, service calls, persistence, and response structures work together.

Implemented Tests

- **test_home_page:** verifies the home page returns HTTP 200 and contains the FitBuddy brand.
- **test_health:** verifies the health endpoint returns HTTP 200 and status='ok'.
- **test_generate_api:** posts a valid user profile and checks for a successful response, user ID, workout plan, and nutrition tip.
- **test_feedback_api:** submits feedback for the generated test user and checks that an updated plan is returned.

Additional Recommended Tests

- Reject age below 13 or above 100.
- Reject weight outside the configured range.
- Reject unsupported intensity or experience values.
- Reject blank/too-short user IDs and feedback.
- Submit feedback for an unknown user and verify HTTP 404.
- Verify deleting a user removes associated plans.
- Verify /api/users returns stored profile fields.
- Verify Gemini fallback output is available when no API key is configured.
- Verify repeated user IDs update the existing profile rather than creating duplicate users.

Manual UI Test Cases

TC-01 Generate plan

Enter valid profile data, submit the form, and verify the result page displays profile details, workout plan, and nutrition/recovery tip.

TC-02 Feedback update

Enter a request such as adding more cardio or another rest day. Verify the page indicates that the plan was updated and displays the revised plan.

TC-03 Admin dashboard

Open /view-all-users and verify the stored profile and plan appear.

TC-04 Delete

Use the dashboard Delete action and verify the user no longer appears.

Acceptance Criteria

- All supplied automated tests pass.
- Invalid input is rejected without creating invalid records.
- Valid generation produces both workout and nutrition outputs.
- Feedback updates an existing plan rather than creating an unrelated user.
- Admin operations do not expose application errors during normal use.
- The app can be demonstrated locally using fallback behavior when Gemini is not configured.

Quick Reference

Test	Expected Result
Home page	200 + FitBuddy text
Health	200 + status ok
Generate API	200 + workout_plan + nutrition_tip
Feedback API	200 + updated_plan
Invalid input	Validation error / rejected request
Unknown feedback user	404

Project basis: This document is derived from the supplied FitBuddy source package, including its README, FastAPI routes, SQLAlchemy models, Jinja2 templates, CSS, and tests.